

SVOBODNY, Russian Federation

WMO: 314450

Lat: 51.4555N

Lon: 128.1070E

Elev: 200

StdP: 98.95

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.5	-35.1	-40.2	0.1	-37.3	-37.9	0.1	-34.8	6.8	-18.1	6.3	-18.4	0.7	310	0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	30.0	20.3	28.1	19.9	26.5	19.2	22.3	27.1	21.4	26.0	20.6	24.8	2.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.9	15.9	24.5	20.0	15.1	23.4	19.1	14.2	22.7	67.0	27.1	63.6	26.3	60.6	24.8	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.5	5.8	-40.5	34.0	3.2	2.6	-42.8	35.8	-44.7	37.3	-46.5	38.8	-48.8	40.7		
(5)			-40.4	24.0	3.2	1.3	-42.7	25.0	-44.5	25.8	-46.3	26.5	-48.6	27.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.2	-23.6	-18.8	-9.1	3.0	11.5	17.9	20.8	18.4	11.4	1.2	-12.9	-23.2
	DBStd	16.67	5.23	5.17	6.22	4.57	4.04	3.37	2.60	3.06	4.06	5.05	6.75	5.43
	HDD10.0	4705	1042	807	591	214	30	1	0	1	32	274	686	1028
	HDD18.3	6917	1300	1040	849	459	216	48	7	36	209	529	936	1287
	CDD10.0	989	0	0	0	5	76	236	336	261	73	2	0	0
	CDD18.3	161	0	0	0	0	3	33	85	38	1	0	0	0
	CDH23.3	1404	0	0	0	8	73	369	651	282	21	0	0	0
	CDH26.7	357	0	0	0	2	15	114	173	52	1	0	0	0
Wind	WSAvg	2.3	2.0	2.2	2.7	3.1	2.8	2.0	1.8	1.9	2.2	2.7	2.3	2.0
Precipitation	PrecAvg	567	7	6	12	27	63	84	139	109	65	30	16	10
	PrecMax	939	27	21	77	113	154	157	304	243	144	81	42	28
	PrecMin	407	1	0	0	1	12	19	46	24	13	6	1	1
	PrecStd	127	6	5	15	26	35	32	75	50	37	18	10	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.8	-2.1	9.2	23.2	28.7	32.7	33.3	30.2	25.6	18.7	5.6	-5.2
		MCWB	-10.5	-4.7	2.7	12.0	15.3	20.0	21.9	21.0	17.7	10.6	2.0	-6.8
	2%	DB	-11.6	-5.3	6.0	17.5	25.3	29.7	30.2	28.0	23.4	14.3	1.6	-9.8
		MCWB	-12.8	-7.3	0.9	8.1	13.7	19.0	21.4	20.5	16.5	7.9	-1.1	-10.9
	5%	DB	-13.8	-8.3	3.1	14.2	22.7	27.2	28.4	26.3	21.1	11.5	-0.6	-12.8
		MCWB	-14.8	-9.9	-0.9	6.5	12.4	18.4	20.9	19.8	14.7	6.3	-2.4	-13.8
	10%	DB	-15.9	-10.8	0.6	11.4	20.1	25.2	26.9	24.7	18.8	9.1	-3.0	-15.3
		MCWB	-16.7	-12.2	-2.3	4.7	11.3	17.5	20.2	19.0	13.6	4.7	-4.8	-16.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-10.2	-4.5	3.3	12.3	16.6	22.2	23.8	23.1	19.2	11.9	2.1	-6.8
		MCDB	-9.0	-2.5	8.6	22.4	25.5	30.2	28.3	27.0	23.4	17.5	5.4	-5.3
	2%	WB	-12.8	-7.3	1.0	8.7	14.8	20.6	22.7	21.9	17.5	8.8	-0.6	-10.8
		MCDB	-11.6	-5.4	5.0	16.1	23.2	27.0	27.5	25.9	21.7	13.1	1.2	-9.7
	5%	WB	-14.8	-9.9	-0.6	6.9	13.5	19.3	21.9	20.8	16.0	6.8	-2.4	-13.7
		MCDB	-13.9	-8.3	2.9	13.4	20.4	25.2	26.6	24.7	19.7	10.9	-0.7	-12.8
	10%	WB	-16.7	-12.1	-2.3	5.2	12.2	18.4	21.2	19.9	14.6	5.1	-4.6	-16.1
		MCDB	-15.9	-10.8	0.6	10.7	18.5	23.6	25.6	23.3	17.8	8.6	-3.1	-15.3
Mean Daily Temperature Range	5% DB	MDBR	11.6	13.3	12.6	12.3	12.5	11.8	10.6	10.8	12.3	11.3	10.8	10.5
		MCDBR	11.5	14.0	13.7	17.7	18.0	16.0	13.7	13.7	15.7	15.3	11.8	10.5
		MCWBR	10.5	12.2	9.4	10.4	8.3	6.9	6.0	6.7	8.6	9.5	9.3	9.6
	5% WB	MCDBR	11.5	14.0	13.0	16.3	14.9	13.3	11.2	10.6	13.3	13.6	11.7	10.4
		MCWBR	10.4	12.1	9.0	9.8	7.6	6.5	5.7	5.7	7.8	8.9	9.3	9.4
Clear-Sky Solar Irradiance	taub	0.264	0.298	0.376	0.553	0.617	0.499	0.480	0.423	0.392	0.388	0.314	0.268	
	taud	2.150	2.075	1.911	1.669	1.620	2.000	2.109	2.259	2.260	2.094	2.120	2.099	
	Ebn at Noon	739	806	792	684	664	766	775	804	784	682	648	653	
	Edh at Noon	73	105	152	221	243	168	148	121	108	104	76	65	
All-Sky Solar Radiation	RadAvg	1.53	2.73	4.38	5.10	5.42	5.99	5.47	4.78	3.74	2.47	1.56	1.18	
	RadStd	0.05	0.08	0.22	0.53	0.42	0.68	0.44	0.32	0.30	0.14	0.08	0.06	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page